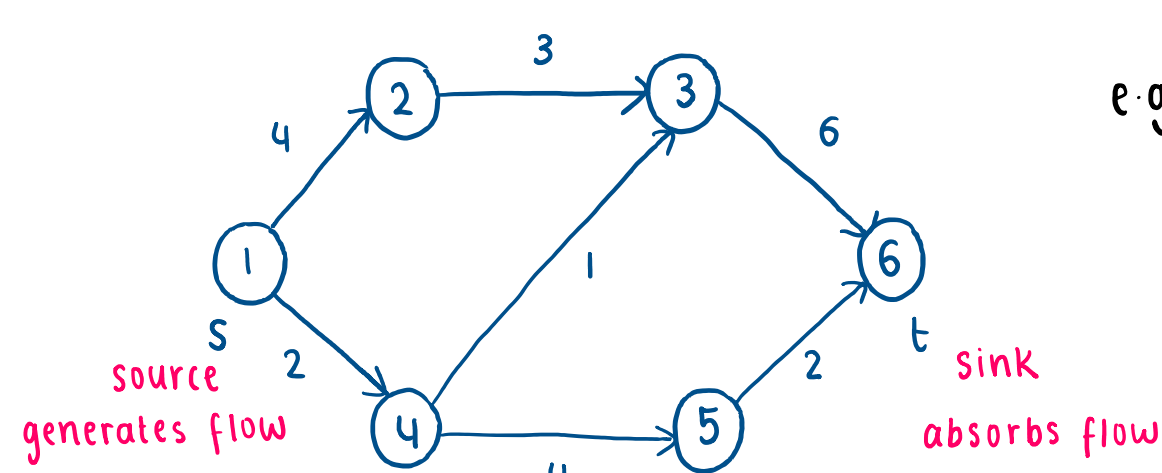


## • Network Flows / Max. Flow Problem

- \* Directed s-t graphs
- \* Capacities on edges



e.g. network routing  
liquid flow

- \* For all internal nodes:

Flow reaching a vertex = Flow leaving a vertex  
(conservation of flow)

i.e. no reservoirs at vertices

OBJECTIVE: maximise flow between s to t

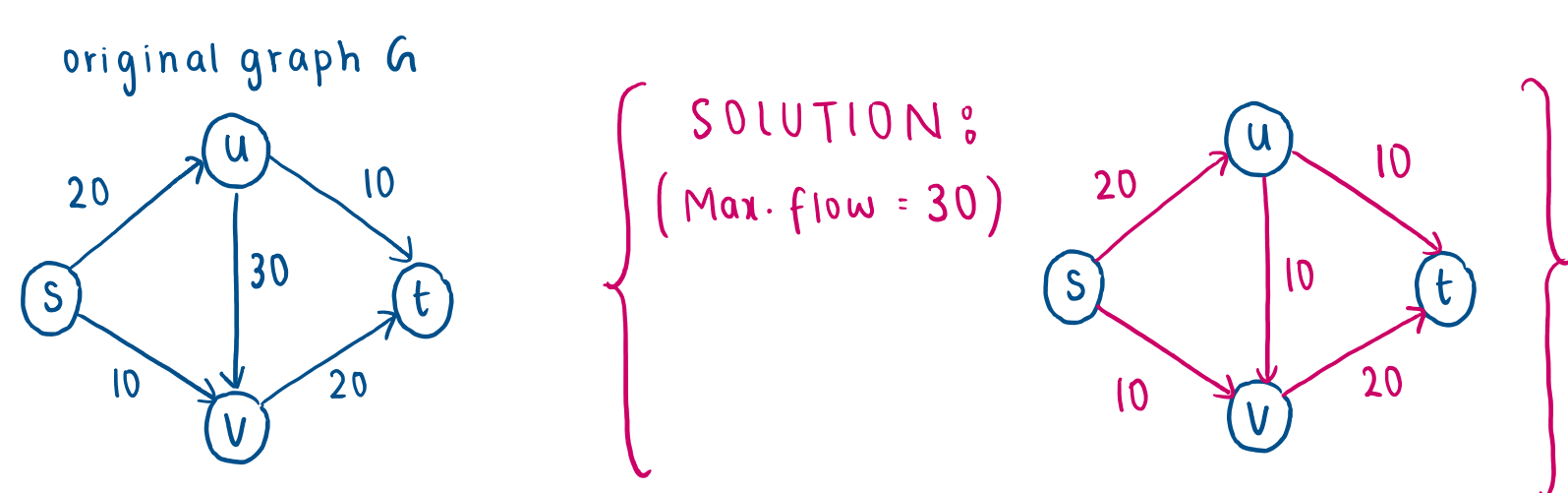
$$\left. \begin{array}{l} \text{Capacities: } c(u \rightarrow v) \\ \text{Flow: } f(u \rightarrow v) \end{array} \right\} \forall u \rightarrow v \in E$$

1.  $f(u \rightarrow v) \leq c(u \rightarrow v)$
2.  $\forall u \in V \setminus \{s, t\}$ :

$$\underbrace{\sum_{\substack{u \in V \\ u \rightarrow v \in E}} f(u \rightarrow v)}_{\text{incoming flow to } v} = \underbrace{\sum_{\substack{w \in V \\ v \rightarrow w \in E}} f(v \rightarrow w)}_{\text{outgoing flow from } v}$$

$$3. \text{ Maximise } \sum_{\substack{w \in V \\ s \rightarrow v \in E}} f(s \rightarrow w) \quad / \quad \sum_{\substack{u \in V \\ u \rightarrow t \in E}} f(u \rightarrow t)$$

### • EXAMPLE:



1. Look for an s-t path  $P$ .

$$P_1 = s \rightarrow u \rightarrow v \rightarrow t$$

augmenting path  
= any s-t path

2. Let bottleneck (min. residual capacity of any edge in  $P$ ) =  $b(P) = \min_{(u \rightarrow v) \in P} \{c(u \rightarrow v)\}$

$$b(P_1) = 20$$

graph with residual capacities

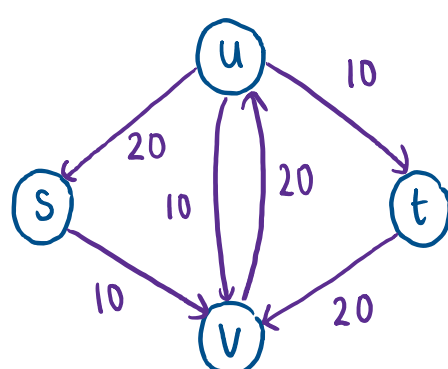
3. Update flow. Compute residual graph.

Residual capacity (after flow  $f$ ):

$$c_f(x \rightarrow y) = \begin{cases} c(x \rightarrow y) - f(x \rightarrow y) & \text{if } x \rightarrow y \in E \\ f(x \rightarrow y) & \text{if } y \rightarrow x \in E \\ 0 & \text{otherwise} \end{cases}$$

edge set of original graph  $G$

$$F = F + b(P_1) \\ F = 0 + 20 = 20$$

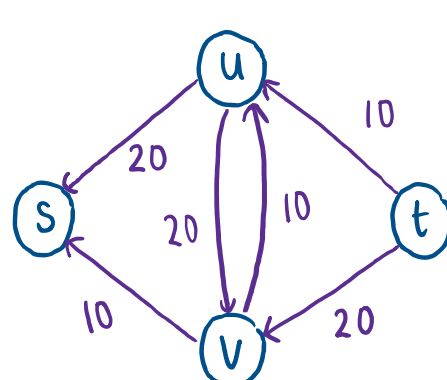


RESIDUAL GRAPH  $G_1$

4. Consider next s-t path.

$$P_2 = s \rightarrow v \rightarrow u \rightarrow t \\ b(P_2) = 10$$

$$F = F + b(P_2) = 20 + 10 = 30$$



RESIDUAL GRAPH  $G_2$

5. No more s-t paths  $\therefore$  algorithm terminates.

Max. flow = 30

### FORD-FULKERSON ALGORITHM

INIT:  $f(e) = 0 \quad \forall e \in G, \quad G_f = G, \quad F = 0$

While there is an s-t path in the residual graph  $G_f$ :

\* Let  $P$  be an s-t path in  $G$ .  $O(m+n)$

\* Find the bottleneck  $b(P)$ .  $O(n)$

\* For each edge  $(u, v) \in P$ :  $O(m+n)$

If  $e = (u, v)$  is a forward edge:

$$f(e) = f(e) + b(P)$$

Else ( $e = (u, v)$  is a backward edge, let  $e = (v, u)$ ):

$$f(e) = f(e) - b(P)$$

\*  $F = F + b(P)$

\* Compute the residual graph  $G_f'$ .  $O(m+n)$

\*  $G_f \leftarrow G_f'$

\* Consider  $e = (u, v)$ .

If  $(u, v) \in E(G)$

$\Rightarrow e$  is a forward edge.

Else  $(v, u) \in E(G)$

$\Rightarrow e$  is a backward edge

$\therefore$  Running time = # iterations  $\times O(m+n)$

Obs: Flow increases in every step by  $\geq 1$  (assuming  $c \in \mathbb{Z}^+$ )

$$\# \text{ iterations} = \min \left\{ \begin{array}{l} \sum_{u: s \rightarrow u \in E} c(s \rightarrow u) \\ \sum_{v: v \rightarrow t \in E} c(v \rightarrow t) \end{array} \right\} \quad \left. \begin{array}{l} \text{termination} \\ \text{condition} \end{array} \right\}$$

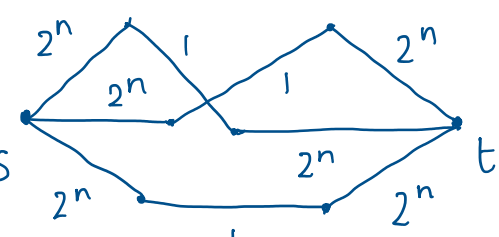
### REPEAT

Find s-t path  $P$ .

Find  $b(P)$ .

Update  $F = F + b(P)$ .

Compute residual graph.



Bound on iterations

$$\leq 3 \cdot 2^n$$

(OVER COUNTING)